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In-Field Evaluation of a Detector-Free Vision Pipeline for Cabbage Crop-Row Guidance

Manh V. Hoang^{†1}, Thang M. Pham¹ and Nam Do¹

¹University of Engineering and Technology, Vietnam National University, Hanoi 10000, Vietnam

[†]Correspondence should be sent to manhhv87@vnu.edu.vn

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Abstract: Vision-only crop-row guidance lets a low-cost field robot steer without RTK-GPS: for every frame the steering controller consumes a heading and a cross-track offset, not a detection score. These are usually produced by a trained row detector, which needs costly field-specific labelling and degrades as the canopy drifts from its training distribution. This work investigates, on a cabbage field robot, whether a detector-free and training-free pipeline of classical components can deliver the same interface. An Excess-Green column-peak tracker confined to a ground safety corridor locates the central row; a robust quadratic fit models its curvature; the heading is the look-ahead tangent and the cross-track its lateral offset; and a temporal stage of a median filter, an exponential moving average, and a physical jump gate stabilises both signals. As nothing is trained, the field runs form a leakage-free test set. Against a 150-frame human-labelled subset the pipeline attains a median heading error of 5.0° and a median cross-track error of about 1 cm, the temporal stage rather than the perception choice being the dominant source of robustness; a residual tail remains where the index locks onto a neighbouring row. Against a trained YOLOv5 detector baseline the median heading gap is only 0.2° and not resolvable at this sample size (95% bootstrap interval $[-0.5, 1.2]$), though the detector stays uniformly ahead in point estimate and decisively better on those wrong-row frames, at zero labelling and GPU cost. The scope is one robot, one crop, and one camera pose.

Keywords— crop-row guidance, vision-based navigation, vegetation index, robust line fitting, temporal filtering, agricultural robot, cabbage.

1. Introduction

Autonomous ground robots are increasingly used for the between-row work in precision agriculture that once required manual scouting and hand-steered passes. Their core requirement is to remain centred on the crop row across varying canopy and growth-stage conditions. A forward-view camera observes the canopy, a perception module localises the central row, and a *guidance line* summarising where the row leads is passed to a steering controller that keeps the vehicle centred [1–3]. On a low-cost robot without RTK-GPS this vision-only guidance line is the sole interface between perception and actuation, and what the controller consumes is a *heading* and a *cross-track offset*, not

a detection score [4, 5].

Most crop-row systems obtain that line from a trained object detector, which raises two difficulties. The detector is optimised for a box-level objective such as mean average precision, not for the heading the controller consumes. It also needs field-specific labelled data that is costly to obtain and prone to leakage between training and test, and it introduces a learned failure surface that is hard to audit in the field [1, 5]. Detector-free, training-free row extraction is itself a classical alternative [6–8]; the open question we address is not *whether* it works but *which* component carries its robustness, and how close it comes to a trained detector when measured against the same ground truth.

We study a deliberately simple pipeline assembled entirely from classical computer vision and evaluate it in the field on a cabbage robot. An Excess-Green vegetation index [9, 10] and a column-peak tracker locate the central row; a robust quadratic fit $x = P(y)$ models its curvature [6, 11]; the heading is taken as the look-ahead tangent to the fitted curve and the cross-track as its lateral offset; and a temporal stage of a sliding-window median, an exponential moving average, and a physical jump gate stabilises both signals over time [12]. The pipeline has no learned weights and trains on nothing, so there is no train/test split to leak and the same code runs unchanged across crops and growth stages. All components are established; our aim is to quantify the guidance accuracy and temporal stability this stock composition achieves in field operation.

This paper makes three contributions. (i) A public forward-view cabbage-robot crop-row guidance dataset, recorded on a working field robot at a camera pose representative of low-cost in-row navigation. (ii) A leakage-free in-field evaluation of the detector-free pipeline against a human-labelled subset, including a comparison with a trained YOLOv5 detector baseline: the complete pipeline trails it by a median heading gap of only 0.2° that is not resolvable at $n = 150$ (95% bootstrap interval $[-0.5, 1.2]$; the gap stays unresolved when the detector is given the same temporal stage),

while the detector remains uniformly ahead in point estimate and decisively better on the wrong-row frames—the measured cost of going detector-free, at zero labelling and GPU cost. (iii) The finding that, across the front ends and indices tested, the temporal stage rather than the perception choice is the main source of heading stability.

The scope is deliberately narrow. We study one robot, one crop, and one camera pose, and we make no claim of generalisation beyond this setting. Section 3 describes the platform, the pipeline, and the evaluation protocol; Section 4 reports the results; and Sections 5 and 6 discuss the findings and conclude.

2. Related work

Four areas bound the present study. The first surveys vision-based crop-row guidance and the shift from classical pipelines to learned ones, and concludes that learned methods currently achieve the highest reported accuracy yet still depend on labelled data and remain exposed to domain shift. The second isolates the classical, handcrafted row-extraction recipe of vegetation index, thresholding, then a Hough or projection search, the family to which a training-free pipeline belongs. The third takes up an argument made with growing frequency in this literature, that crop-row methods ought to be measured in *navigation units* rather than detection mAP. Temporal filtering and path-tracking form the fourth, the back-end machinery that converts noisy per-frame geometry into a usable heading. Against this background sits a detector-free, temporally-robust pipeline evaluated on in-field cabbage video.

2.1. Vision-based crop-row guidance: classical versus learned

Steering a robot or tractor along a crop row from a forward-view camera is a long-standing task, and vision-guided field machinery goes back several decades [3, 13]. Early systems recover row geometry without any learned segmentation. Intensity-based crop-row determination by image analysis [5] and global energy minimisation over candidate row models [14] are representative of the classical, hand-engineered line of work.

Perception in this area has since moved decisively toward learned models. The de Silva et al. line of work segments the crop/soil mask with a network and recovers a centreline through a geometric scan, validated across many field conditions and growth stages [1, 2]. Detect-then-fit systems instead localise row patches or plant clusters with an object detector and fit a guidance line through the centroids, as in the YOLOX-based maize-row system of Gong et al. [15]. A further group regresses the row directly: RowDetr predicts each row as a polynomial [16], while a row-column attention model recasts the whole pipeline as end-to-end attention [17]. The row-line formulation itself descends from lane detection. There, Ultra-Fast structure-aware lane detection casts the problem as row-anchor classification [18], LaneATT performs anchor-based real-time detection [19], and Van Gansbeke et al. fit a lane as a differentiable least-squares line through detected points [20].

Learned methods currently achieve the highest reported accuracy, but at the cost of supervision. Each requires pixel- or row-level labels for the target crop, growth stage, and camera geometry, and its accuracy is assured only on distributions that resemble the training set. Degradation under changes of field, crop, and acquisition condition recurs across this family. For a single robot, a single crop, and a fixed camera pose, a training-free pipeline that needs no labelled training data is a label-free alternative warranting in-field validation against navigation-unit ground truth. The classical family that supplies such a pipeline is examined in the following subsection, and its robustness across a

long real run remains unaddressed by the learned literature.

2.2. Classical, handcrafted row extraction

The pipeline studied here belongs to the classical family that predates learned segmentation, and its components serve as stock parts rather than new contributions. Three stages make up the standard recipe. A vegetation index first separates plant from soil in colour space; the Excess-Green index and related colour indices for plant/weed discrimination under varied soil, residue, and lighting were established by Woebbecke et al. [9] and verified for automated crop imaging by Meyer and Neto [10]. The index image is then binarised, classically with Otsu’s histogram threshold [21]. Row geometry is finally recovered from the resulting mask by a global search, either a Hough transform for real-time plant-row tracking [6] or the column projection and scan-line peaks used in vision-based row-following and row-detection systems for field machinery and sugar beet [7, 8].

Maturity, speed, and the absence of labels make these components a natural choice. The present pipeline instantiates the recipe with an Excess-Green column-peak central-row tracker. A robust quadratic fit $x = P(y)$ by Huber IRLS ($\delta = 1.345$, scale 1.4826 MAD [11]; RANSAC [22] is the standard robust alternative) replaces the single straight-line Hough or projection model, so that the guidance line can bow in the far field, and heading θ follows as the look-ahead tangent at a chosen row position, with cross-track offset e taken at the same point. The classical literature does not settle how such a single-frame extractor behaves over a long, real, forward-view field run across varying vegetation density, nor how its per-frame output should be stabilised in time. Both questions are addressed in the following subsection and in the experiments reported here.

2.3. Evaluation in navigation units

A recurring theme in recent crop-row work holds that detection mAP is the wrong yardstick for a guidance system. A detector can score well on mAP and still yield a poor guidance line, because what matters downstream is the geometry of the recovered row rather than the quality of individual boxes. The de Silva et al. benchmarks therefore report cross-track and heading error in degrees and pixels [1, 2], and the maize-row detect-then-fit system reports the angular error of the fitted centreline rather than box mAP [15]. Heading error and cross-track error, measured against a human-labelled guidance line, are the quantities that matter for guidance.

Navigation-unit evaluation needs a ground-truth line, which in turn needs human labelling and a study of inter-annotator agreement. Absent that line, an extractor can report how often it emitted a line but cannot report whether the emitted line was correct. The accuracy claim against a human ground truth is reported here on a labelled subset; the inter-annotator agreement that would set a floor on it is left to future work. The camera calibration that expresses cross-track in centimetres is built from the robot’s measured camera pose (Section 3.1).

2.4. Temporal filtering and path tracking

Per-frame row geometry from any single-frame extractor is noisy, so practical guidance systems stabilise it over time. The canonical tool is the Kalman filter, which fuses a motion model with noisy measurements for recursive state estimation [12] and is widely used in lane- and row-tracking work. Once a stabilised heading and offset are available, classical path-tracking controllers convert them into steering. Pure pursuit follows a look-ahead point on the path [23], the Stanley controller couples heading and cross-track error for autonomous ground vehicles [24], and the kinematic bicycle model supplies the standard vehicle abstraction these controllers assume [25].

In the crop-row literature this temporal back end, though standard, is seldom isolated as a controlled ablation against the perception choice, so its relative contribution to per-frame stability is rarely quantified. A transparent temporal stage, kept in front of a simple stock extractor, occupies the role that learned pipelines fill implicitly, and this paper measures its contribution directly. The substantive robustness questions, namely heading error against a human ground truth on the frames where the raw extractor diverges, and the selectivity of the gate, are answered against the labelled subset in Section 4.4; the fit-arm comparison against RANSAC and equal-weight least-squares fits is reported there, and the comparison against the platform’s trained detector is deferred to that baseline run.

2.5. Positioning of the present study

Where the literature trends toward learned perception that needs labels and is exposed to domain shift, the gap left open is a detector-free and training-free pipeline assembled from stock classical parts: an Excess-Green column-peak tracker [9, 10, 21], a robust quadratic fit $x = P(y)$ by Huber IRLS [11], a look-ahead tangent heading, and a median/EMA/jump-gate temporal stage [12]. All components are established. Positioned against prior art, the open question is whether such a pipeline holds up on real forward-view cabbage field video and can be validated in navigation units against a human ground truth, a question the remaining sections take up.

3. Materials and methods

This is a validation and deployment contribution, not a new method: the pipeline is assembled entirely from established, stock cv2/numpy components with no neural detector and no training, so every frame is a test frame and no data can leak into the evaluation. This section describes the

platform, the pipeline, and the evaluation protocol. The accuracy results are measured against a human-labelled subset of 150 frames – row emission, cross-track magnitude, heading and cross-track error, the temporal ablation, and the fit-arm and trained-detector baselines – by a single annotator, so an inter-annotator agreement floor on the smallest meaningful error is not established and is left to future work.

3.1. Robot platform and field data

The data were collected from a single ground robot (Figure 1), a four-wheel straddle platform that drives along a crop row with a forward-view camera mounted to look down the inter-row corridor. The field runs were recorded as the robot drove along the cabbage rows at a nominal forward speed of approximately 0.14 m s^{-1} . The platform pairs an embedded CPU-only computer and a forward-view USB webcam at 640×480 , here a Logitech HD C270 webcam and an embedded board of the NVIDIA Jetson Nano / Raspberry Pi 4 class. The detector-free pipeline of this paper runs on its CPU with no neural accelerator (Figure 2). This robot shares its camera model with a sibling robot whose camera was calibrated from a standard checkerboard procedure; the intrinsics and lens distortion from that calibration are reused here, as they are a property of the camera and lens rather than of the mounting. The extrinsics, by contrast, are taken from this robot’s own measured installation: the camera sits on the robot centreline 65 cm above the ground with its optical axis tilted 55° from the vertical (equivalently 35° below the horizontal). From these intrinsics and measured extrinsics an image-to-ground homography is constructed under a flat-ground assumption (world origin on the centreline beneath the camera) and used to express the near-row cross-track in centimetres (Section 3.4). The residual assumptions are the planar ground and the reuse of the camera intrinsics, so the cross-track is also reported in pixels and as a fraction of the half-image width. The centimetre scale is sensitive to the measured pose: perturbing the height by $\pm 5 \text{ cm}$ and



Figure 1: The cabbage-field robot during a field-test session: a four-wheel straddle platform carrying a forward-view camera, driving along the cabbage rows. The guidance pipeline runs on the platform’s onboard CPU; closing the steering loop from the pipeline’s output is left to future work.

the tilt by $\pm 5^\circ$ moves the near-row pixel-to-centimetre factor enough to shift the reported median cross-track over roughly $[0.9, 1.3]$ cm, so the centimetre figures are indicative and the pixel error is treated as the primary cross-track metric. The scope is fixed to one robot, one crop (cabbage), and one camera pose. The validation reported here is narrow by design and at a single low frame rate, with no claim of generalisation beyond this configuration. Integrating the guidance into the steering loop and measuring the resulting closed-loop tracking is left to future work.

The full sensing-to-actuation hardware path appears in Figure 3. The forward-view camera streams over USB to the onboard computer, which runs the detector-free guidance pipeline and emits a heading and a cross-track offset. A microcontroller and stepper drives turn the wheels. During



Figure 2: Onboard camera and compute. The forward-view camera and the embedded computer are carried on the four-wheel straddle frame that spans the crop bed. The bed shown here is mixed, with cabbage interplanted among other leafy greens, as also reflected in the field runs (Section 4.1).

the recorded runs the loop was closed by the platform’s existing detector-based controller; for the proposed pipeline only the perception path was run and logged, from camera to guidance output, and closing the loop from that output to the drive is marked as future work.

Two real robot runs recorded on cabbage plots form the frame set: run 1 (approximately 20 minutes) and run 2 (approximately 14 minutes). Frames are sampled at 1 frame per second (1 fps) and screened to the frames showing the cabbage navigation row through the look-ahead region, removing stretches dominated by adjacent leafy-green beds or by occluding operators. This yields 1482 forward-view cabbage frames in total (812 from run 1 and 670 from run 2), all at 640×480 . The full set is treated as a pure test set. Because the pipeline is training-free there is no training/validation split and no possibility of memorisation. To probe robustness to crop growth stage and canopy

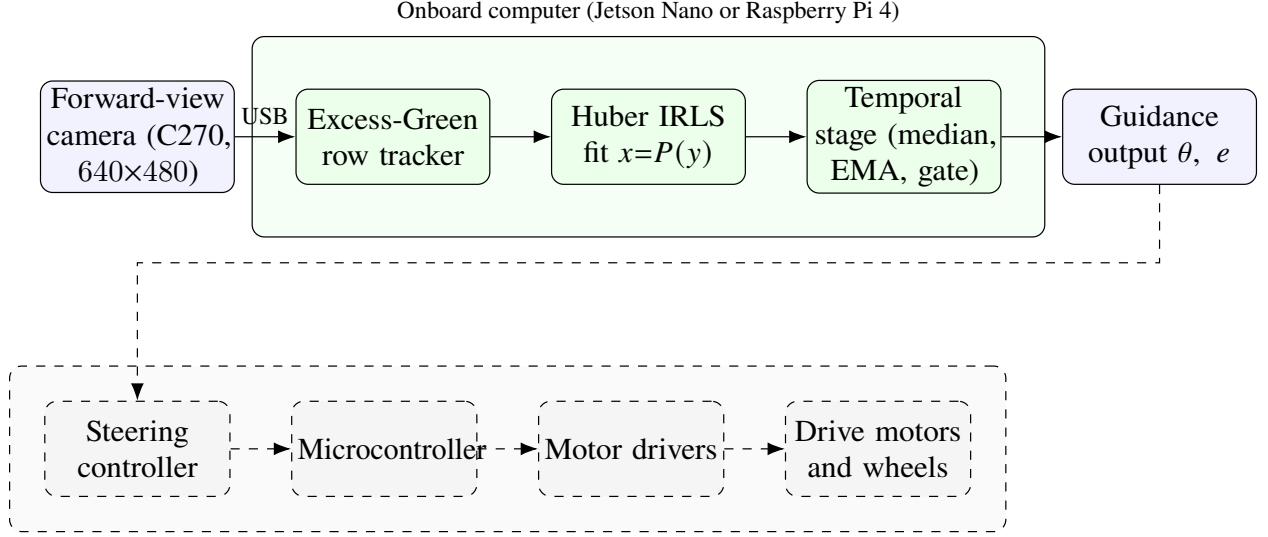


Figure 3: Hardware and data path of the cabbage robot. The solid blocks, sensing and the onboard detector-free guidance pipeline, were run and logged in this study; the dashed group (steering controller, microcontroller, stepper drivers, and motors) was not exercised here and shows the intended closed loop.

density the frames are stratified into vegetation tertiles (low, mid, and high Excess-Green coverage), giving three roughly equal strata of 491, 487, and 504 frames. The dataset will be released publicly as the first real forward-view cabbage robot crop-row corpus, and a Data and Code Availability statement accompanies the manuscript.

3.2. Central-row tracking via the Excess-Green index

Each frame is reduced to a single-channel vegetation map using the Excess-Green index of Woebbecke *et al.* [9]. From the normalised RGB channels (r, g, b) the index is

$$\text{ExG} = 2g - r - b, \quad (1)$$

which emphasises live green canopy against soil and residue and is a standard, illumination-tolerant choice for in-field crop/soil separation [10]. The map is thresholded into a binary vegetation mask by Otsu’s method [21], which selects the foreground/background split automatically for each frame and so adapts to changing lighting and canopy density without a hand-tuned constant.

Crop rows viewed head-on appear as near-vertical green bands, and the central navigation row is the band the robot straddles. Tracking it without a detector relies on a per-row column profile. For each image row y the pipeline sums (or smooth-and-sums) the masked ExG response across columns and takes the column of peak vegetation density as the candidate row centre $x^*(y)$ at that height. Scanning y from the image bottom upward yields a set of central-row anchor points $\{(x^*(y), y)\}$ that follow the near-vertical green band. To bias the tracker toward the row the robot is actually in rather than an adjacent one, the column search is restricted to a *virtual safety corridor*: a lateral band about the robot centreline on the ground, projected into the image per row through the calibration of Section 3.1. The platform’s straddle chassis spans a single bed of half-width 220 mm, so the neighbour rows sit near or just inside that width. Because a vegetation index cannot tell the central crop from a neighbouring crop, the seed band is set slightly tighter, at half-width ± 180 mm: when the central row is momentarily gapped, a ± 220 mm band would still admit the neighbour at the chassis edge and the (memoryless) front end would re-seed onto it every frame of the gap, whereas the ± 180 mm band excludes it while still admitting the central row’s lateral wander. The corridor is applied when seeding the track at the near row, where the rows are most separated in the image, so that the track starts on the central row and follows it upward. The output of this stage is the ordered set of central-row anchors passed to the line fit. Emitting an anchor set is a self-consistency property and not a correctness guarantee: a peak is found in essentially every frame, but whether the chosen column corresponds to the true central row is precisely what the ground-truth evaluation of Section 3.7 is designed to measure.

3.3. Robust quadratic guidance-curve fitting

The guidance line is parameterised as image column x as a function of image row y ,

$$x = P(y) = c_2 y^2 + c_1 y + c_0, \quad (2)$$

a quadratic in y . This orientation is chosen for a reason. Crop rows viewed head-on are near-vertical, so a conventional $y = mx + c$ fit becomes ill-conditioned as the slope diverges; expressing x as a function of y removes this vertical-line degeneracy [5]. The quadratic term c_2 captures the gentle far-field bowing of a row under perspective and mild row curvature, and reduces to the straight-line case $x = c_1 y + c_0$ in the limit $c_2 \rightarrow 0$. The instantaneous local heading at any height follows from the slope $P'(y) = 2c_2 y + c_1$,

$$\theta(y) = \arctan(P'(y)), \quad (3)$$

reported in degrees from the vertical as the controller-independent navigation quantity; its evaluation at a look-ahead height is detailed in Section 3.4.

Given the n central-row anchors $\{(x_i, y_i)\}$ from Section 3.2, the coefficients $\mathbf{c} = (c_2, c_1, c_0)$ are obtained by a standard robust quadratic fit $x = P(y)$ by Huber iteratively-reweighted-least-squares (IRLS) regression of x on y [11], minimising

$$\hat{\mathbf{c}} = \arg \min_{\mathbf{c}} \sum_{i=1}^n \rho_{\delta}(x_i - P(y_i)), \quad \rho_{\delta}(r) = \begin{cases} \frac{1}{2}r^2, & |r| \leq \delta, \\ \delta(|r| - \frac{1}{2}\delta), & |r| > \delta. \end{cases} \quad (4)$$

The coefficients are estimated by standard Huber IRLS with $\delta = 1.345$ and a MAD scale $\hat{\sigma} =$

243 1.4826 $\text{median}_i |r_i|$, with a small ridge for conditioning, giving the Huber influence weights

$$w_i = \min\left(1, \frac{\delta \hat{\sigma}}{|r_i|}\right), \quad \delta = 1.345. \quad (5)$$

244 The constant $\delta = 1.345$ is the conventional Huber tuning that retains $\approx 95\%$ efficiency under
245 Gaussian noise while bounding the influence of outliers, which here are spurious anchors from gaps,
246 weeds, or an adjacent row.

247 Two standard alternatives serve as comparison fits over the same anchor set, evaluated against
248 the pipeline’s Huber IRLS fit: equal-weight least squares (the non-robust polynomial fit) and
249 RANSAC consensus fitting [22], a strong robust baseline. The accuracy of the Huber fit relative to
250 these baselines against human ground truth is reported in Section 4.4: the Huber and equal-weight
251 least-squares fits are statistically indistinguishable and both ahead of RANSAC, so the fit choice is
252 secondary on the corridor-restricted anchors.

253 3.4. Heading and cross-track geometry

254 A controller steers not to the row directly beneath the robot but to where the row is heading some
255 distance ahead. The guidance geometry is therefore evaluated at a fixed look-ahead image row y_{la} in
256 the upper portion of the frame, mirroring the look-ahead point of pure-pursuit and Stanley-type path
257 trackers [23–25]. Because the curve $P(y)$ bows in the far field, the relevant heading is the tangent
258 to the curve at the look-ahead height, not a chord to the image bottom. The reported heading is
259 therefore the look-ahead tangent

$$\theta = \arctan(P'(y_{\text{la}})) = \arctan(2c_2 y_{\text{la}} + c_1), \quad (6)$$

the slope of Equation (3) evaluated at y_{la} , in degrees from the vertical. Using the tangent rather than a global slope aligns the reported heading with the far-field row direction the robot must follow, and avoids the bias a straight-line fit would incur on a bowed row.

The lateral cross-track offset e is the signed horizontal distance between the robot’s reference column, taken as the image vertical centreline $x_c = W/2$ for a forward-view camera with $W = 640$, and the guidance line evaluated at the look-ahead row,

$$e = P(y_{\text{la}}) - x_c, \quad (7)$$

in pixels. The offset e is additionally reported normalised by the half-image width $W/2$ (that is, as a fraction of the half-frame), so that it is dimensionless and comparable across mountings. Conversion to centimetres requires a metric ground anchor, namely an image-to-ground homography under a flat-ground assumption. This anchor is the homography of Section 3.1, built from the robot’s measured camera height and tilt; the centimetre cross-track magnitude follows directly, while e is in all cases also reported in pixels and in % of half-image width.

3.5. Temporal filtering

The per-frame heading θ and offset e are smoothed over time by three stock, lightweight operators applied in sequence. Together they form the temporal stage, which is the primary source of robustness. The chain is far simpler than a full state estimator such as a Kalman filter [12]; its complexity is matched to the 1 fps operating regime.

The first operator passes the raw heading stream through a short sliding-window median filter. The median is robust to isolated single-frame outliers, such as a momentary wrong-row peak or a gap-induced bad fit, without the lag or overshoot of a mean filter. The median-filtered heading is

then low-pass filtered by an exponential moving average,

$$\hat{\theta}_t = \beta \hat{\theta}_{t-1} + (1 - \beta) \tilde{\theta}_t, \quad (8)$$

where $\tilde{\theta}_t$ is the median-filtered heading at frame t and $\beta \in [0, 1)$ sets the smoothing horizon. The exponential moving average suppresses high-frequency jitter while remaining causal and $O(1)$ in memory. The offset e is fused with the identical median→EMA chain.

A frame-to-frame jump gate completes the chain, rejecting any update whose heading change exceeds a physical bound $\Delta\theta_{\max}$. When $|\tilde{\theta}_t - \hat{\theta}_{t-1}| > \Delta\theta_{\max}$ the update is gated and the last fused heading is held rather than applied. The bound $\Delta\theta_{\max}$ is set from the robot’s physical heading-rate limit, independently of any evaluation threshold. A ground robot driving a row cannot reorient faster than some maximum yaw rate $\dot{\psi}_{\max}$ (deg s^{-1}), so over one inter-frame interval Δt the largest kinematically plausible heading change is

$$\Delta\theta_{\max} = \dot{\psi}_{\max} \Delta t, \quad \Delta t = \frac{1}{1 \text{ fps}} = 1 \text{ s}. \quad (9)$$

Any apparent inter-frame heading change larger than $\Delta\theta_{\max}$ cannot be the vehicle physically turning; it has to be a perception artefact, such as a jump onto an adjacent row, and is gated. Here $\dot{\psi}_{\max} = 8^\circ \text{ s}^{-1}$, giving $\Delta\theta_{\max} = 8^\circ$ at 1 fps. The value $\dot{\psi}_{\max}$ is fixed from the platform’s turning specification, and hence $\Delta\theta_{\max}$, before any error metric is examined, so the gate threshold is not tuned to the evaluation. The same physical reasoning bounds the per-frame change in e , here at 130 px.

A reduction in large per-frame jumps is not the measure of the gate’s effect, since the gate rejects large jumps by definition and any “before/after” jump count is therefore arithmetic rather than evidence of correctness. Raw-versus-fused jump counts are reported descriptively in Section 4,



Figure 4: Detector-free guidance on a real forward-view cabbage frame. *Left*: input frame (640×480). *Right*: the Excess-Green column-peak track (orange points), the robust Huber IRLS quadratic guidance curve $x = P(y)$ (green), the look-ahead tangent that defines the steering heading θ (red), and the near-row cross-track offset e from the image centre (blue); the reported θ and e are overlaid. No detector and no training are involved.

labelled as such, and are excluded from the robustness results. The substantive robustness claim, that fusion lowers heading error against human ground truth on the frames the gate suppresses while the gate fires selectively on poor frames, is evaluated against the labelled subset in Section 4.4.

All numeric settings are fixed across both runs and listed in Table 1 for reproducibility. Of these, only the ± 180 mm seed-corridor width is selected with reference to the labelled subset, by the width sweep reported in Section 4.4; it is therefore the one parameter exposed to the evaluation set, and choosing it on a disjoint validation set is left to future work. The jump-gate threshold is a physical kinematic bound fixed before any error metric, and the remaining settings are geometric bounds or standard defaults.

Figure 4 shows the complete pipeline applied to a single forward-view cabbage frame, from the Excess-Green track through the robust curve to the look-ahead tangent and the reported heading and cross-track.

Table 1: Pipeline hyperparameters, fixed across both runs. H, W are the image height and width (480×640). Only the seed-corridor width is selected with reference to the labelled subset (Section 4.4); the rest are physical, geometric, or standard defaults.

Stage	Parameter	Value
Front end	vegetation index	Excess-Green
	scan bands	9
	band half-height	$0.04 H$
	column smoothing	21-px boxcar
	peak threshold	mean + 0.3 std of the column profile
	per-band track tolerance	$0.18 W$
	seed safety corridor	± 180 mm (ground), projected per row
Curve fit	model / degree	quadratic $x = P(y)$, degree 2
	fit band	$[0.45, 0.98] H$
	robust loss	Huber, $\delta = 1.345$, 5 IRLS iterations
	look-ahead row (heading)	$0.65 H$
	cross-track row	$0.92 H$
Temporal	median window	15 frames
	EMA factor β	0.4
	heading jump gate $\Delta\theta_{\max}$	$8^\circ/\text{frame}$ ($\dot{\psi}_{\max} = 8^\circ \text{ s}^{-1}$)
	cross-track jump gate	130 px/frame
	warm-up	3 frames
Calibration	camera height / tilt	650 mm / 55° from vertical
	intrinsics	reused C270 checkerboard K, D
Baselines	RANSAC	200 iterations, 8-px inlier threshold
	Kalman (const.-velocity)	process var $q = 0.5$, measurement var $r = 9.0$

3.6. Ground-truth annotation protocol

The ground-truth (GT) guidance line used for evaluation is human-annotated and is therefore independent of the perception pipeline under test. For each labelled cabbage frame an annotator marks the central navigation row directly, following *convention A*: the line is labelled along the row the robot is driving, traced through the cabbage canopy from the image bottom-centre up the

inter-row corridor. The marked points are fit in the same $x = P(y)$ parameterisation of Equation (2), and the GT heading is the look-ahead tangent of Equation (6) evaluated at the same y_{la} as the system output, so that predicted and GT headings are directly comparable. A subset of 150 frames was labelled by a single annotator, stratified across the two runs and the vegetation tertiles. Quantifying the reliability of the protocol would require a second annotator: an inter-annotator agreement, the median absolute heading difference in degrees and cross-track in pixels, would set a floor on the smallest meaningful error, and that study is left to future work. Because the GT line is marked by hand on the image rather than derived from the tracker’s anchors, it is independent of the Excess-Green front end and provides a fair external reference.

3.7. Evaluation metrics and statistics

The primary metric is heading error in degrees, the absolute difference between the system’s look-ahead tangent heading and the human-GT heading, $|\theta - \theta^{gt}|$, which is controller-independent. The secondary metric is cross-track error in pixels (and in % of half-image width), the lateral offset between the predicted and GT guidance lines at the look-ahead row y_{la} (Equation (7)). Centimetre units are reported through the image-to-ground homography of Section 3.1, built from the robot’s measured camera pose; the heading and cross-track errors against the labelled subset are reported in degrees, pixels, and centimetres in Section 4.4.

As a basic sanity check the fraction of frames for which the tracker emits a guidance line at all is reported as a self-consistency measure, both overall and broken down by vegetation tertile, in Section 4.2. Here “row found” means a line was emitted, not that the emitted line is the correct central row; whether the emitted line is correct is measured by the heading and cross-track error against human GT in Section 4.4.

Consistent with Section 3.5, the robustness of the temporal stage is reported as (i) heading error

versus human GT on the gate-suppressed frames, raw versus fused, and (ii) the selectivity of the gate, that is, whether the frames it holds carry larger raw errors than the frames it passes; both are reported in Section 4.4. The raw-versus-fused jump count is excluded from the robustness results. The Huber-fit-versus-baseline comparison (equal-LS, RANSAC) is reported in Section 4.4; the trained-detector baseline is reported there too.

Per-frame errors are written to one CSV row per frame so that the detector-free pipeline and its baselines and ablations are compared paired on identical frames. Because per-frame errors are not normally distributed, comparisons are reported as error distributions (median, mean, and 90th percentile) and as paired per-frame differences rather than as p -values. For the load-bearing comparisons the median paired difference is accompanied by a 95% confidence interval from a run-clustered bootstrap (10,000 resamples, frames resampled within each run so that the within-run correlation of 1 fps frames is not counted as independent evidence); because the labelled frames are a sparse, evenly spaced subset, the run-clustered interval matches the frame-level one to within rounding, indicating the precision is not inflated by temporal autocorrelation. A difference whose interval covers zero is reported as not resolvable at this sample size rather than as a ranking, and the point estimate is reported alongside so that an unresolved difference is not read as demonstrated equivalence. This is the case for the Huber fit against equal-weight least squares (median paired difference 0°) and, notably, for the proposed pipeline against the trained YOLOv5 detector baseline (median 0.2° , interval $[-0.5, 1.2]^\circ$; still covering zero when the detector is also given the temporal stage, 0.5° , $[-0.4, 1.2]^\circ$).

4. Results

This section reports what two real cabbage-field robot runs establish, separating what those runs *measure* from what still depends on a larger or independent labelling effort. The pipeline under test is the detector-free and training-free guidance stack of Section 3: an Excess-Green index [9, 10] column-peak central-row tracker restricted to a fixed ground safety corridor, a robust quadratic fit $x = P(y)$ by Huber IRLS [11], a look-ahead tangent that defines the heading θ together with the cross-track offset e , and a temporal stage of a sliding-window median, an exponential moving average, and a physical max-jump gate applied to both signals. The system is never trained, so the two runs serve as a pure test set; no parameters are fitted on these frames, which removes any path for leakage.

One distinction governs the numbers that follow. Self-consistency is not correctness: the row-found rate of Section 4.2 records only that the tracker *emitted* a guidance line, and it carries no claim about whether that line is the correct central row. Likewise, the post-fusion jump counts of Section 4.3 fall to near zero as a direct effect of the gate threshold and are reported as descriptive context, not as a robustness result. The correctness claim is settled separately in Section 4.4 as error against a human labelling of the central row, including on the frames the gate suppresses. Cross-track is reported in pixels, as a percentage of the half-image width, and in centimetres through the image-to-ground homography of Section 3.1, built from the robot’s measured camera height and tilt.

4.1. Dataset

The evaluation data are two forward-view video runs recorded by the onboard camera of a cabbage field robot, sampled at 1 fps and screened to 1482 forward-view cabbage still frames at 640×480 .

The two runs cover different sections of the field at different times of day, so vegetation density and lighting differ between them. Table 2 summarises the runs. The setting is one robot, one crop, and one camera pose, and the dataset supports no claim of coverage beyond this narrow operating envelope.

Table 2: Field dataset: two real forward-view cabbage robot runs used as a pure test set (no training, so no leakage). Frames are sampled at 1 fps.

Run	Duration	Frames	Rate (fps)	Resolution	Camera
Run 1	~20 min	812	1	640 × 480	forward-view
Run 2	~14 min	670	1	640 × 480	forward-view
Total	~34 min	1482	1	640 × 480	forward-view

4.2. Row-emission reliability

We first assess whether the perception front end emits a guidance line on every frame. Across all 1482 frames the Excess-Green column-peak tracker emitted a guidance line in 99.9% of frames (1481/1482); a single frame produced no peak inside the seed corridor and was abstained on. To test whether this figure conceals a silent failure mode in sparse or dense canopy, we split the frames into vegetation tertiles by per-frame Excess-Green coverage (low, mid, high). Table 3 shows the emit rate is 100% in the low and mid tertiles and 99.8% in the high tertile. This stratification is partly tautological by construction: the tracker locates the strongest Excess-Green column within the safety corridor, so a peak almost always exists whenever any green is present. The result therefore shows only that the tracker essentially never aborts; whether the canopy density leaves the emitted line on the correct row is the separate accuracy question of Section 4.4.

Mapping the fused near-row pixel offset through the image-to-ground homography (Section 3.1) describes the magnitude of the lateral offset in metric units. After the temporal stage the central row

Table 3: Reliability (self-consistency, *not* correctness): fraction of frames in which the tracker emitted a guidance line, overall and by vegetation tertile (Excess-Green coverage). The rate means a line was *emitted*, not that it is the correct central row.

Stratum	Frames	Row emitted	Rate
All frames	1482	1481	99.9%
Vegetation: low (T1)	491	491	100%
Vegetation: mid (T2)	487	487	100%
Vegetation: high (T3)	504	503	99.8%

sat a median of 1.7 cm from the image centre across the runs (mean 3.4 cm, 90th percentile 10.8 cm). These figures characterise the *magnitude* of the lateral offset along the recorded trajectory; they summarise where the row sat in the frame rather than the accuracy of the proposed estimator, which is reported against ground truth in Section 4.4.

4.3. Temporal stability

The temporal stage, a sliding-window median followed by an exponential moving average and a physical max-jump gate in the spirit of standard recursive smoothing [12], exists to suppress single-frame excursions that a controller should not act on. Table 4 reports how many per-frame heading jumps exceed 10° , 15° , and 20° in the raw per-frame tracker output and after the temporal stage. The per-frame jump is the change in heading from the previous frame and is undefined for the first frame of each run, so the denominators are the 810 and 669 logged transitions.

These counts are descriptive, since the gate suppresses the jumps by design: the post-fusion counts fall to near zero because the max-jump gate rejects exactly these large frame-to-frame changes, which is an arithmetic property of the gate threshold rather than an independent measurement of correctness. Whether the temporal stage actually helps, namely whether the fused estimate is closer to human ground truth than the raw estimate on the suppressed frames and how selectively the gate

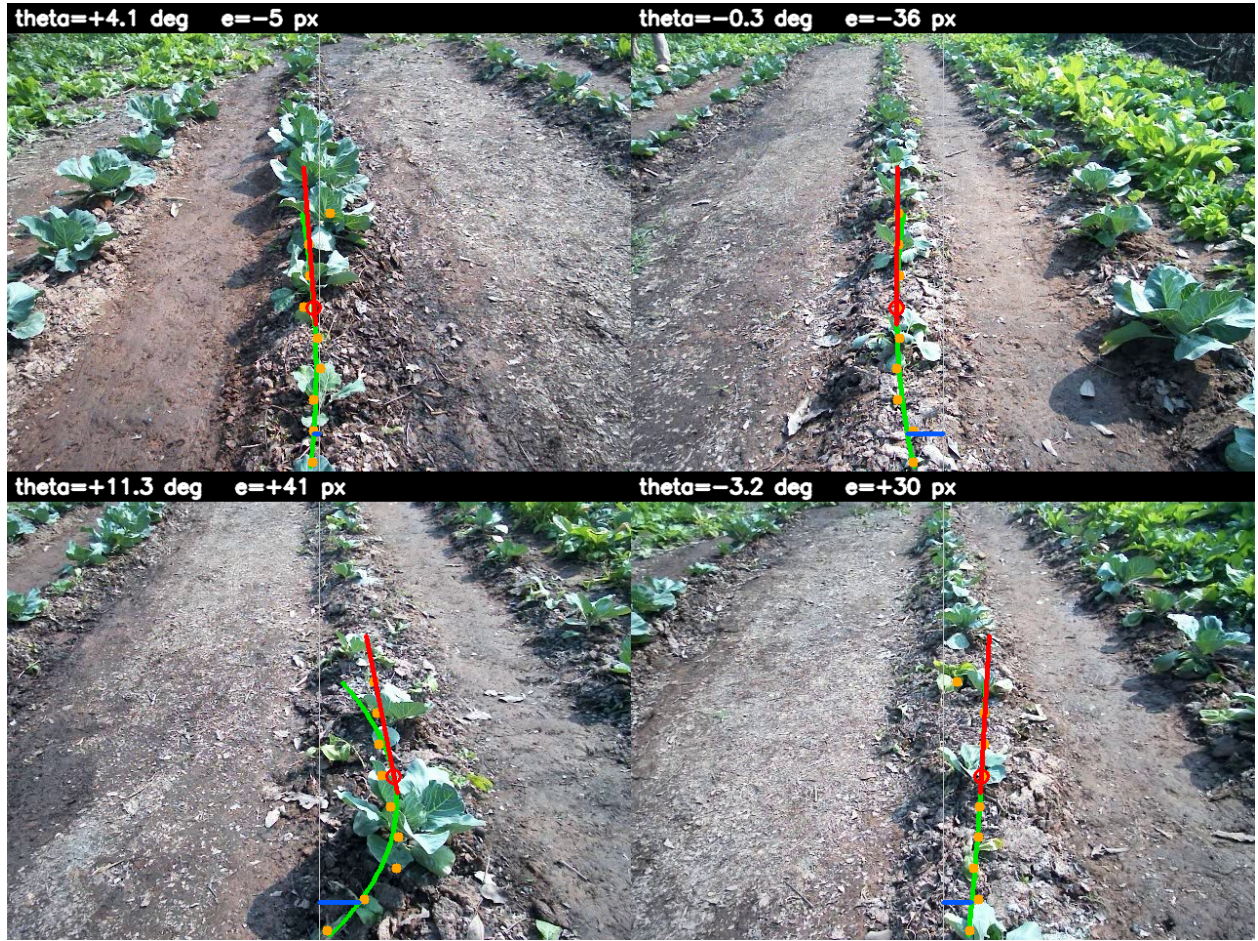


Figure 5: Qualitative behaviour of the detector-free estimator on four real forward-view *cabbage* frames spanning a range of canopy density (each panel overlays the Excess-Green column-peak track in orange, the robust quadratic guidance curve $x = P(y)$ in green, the look-ahead tangent/heading in red, and the near-row cross-track in blue; the reported θ and e are printed). The tracker follows the central cabbage row from densely planted rows through sparse, gapped stretches to young transplants. These panels are illustrative, not a correctness measurement (cf. Section 4.4).

414 fires, is the ablation reported in Section 4.4.

Table 4: Per-frame heading-jump counts (change from the previous frame) exceeding $10^\circ/15^\circ/20^\circ$, raw versus after the temporal stage, as exact counts on the logged frame-to-frame transitions. The post-fusion reduction follows arithmetically from the max-jump gate and is not a correctness result; the correctness test on these frames is the ablation of Section 4.4.

Run / stage	$> 10^\circ$	$> 15^\circ$	$> 20^\circ$
Run 1 raw ($n = 810$)	114	75	45
Run 1 temporal stage	0	0	0
Run 2 raw ($n = 669$)	189	125	85
Run 2 temporal stage	2	0	0

4.4. Accuracy and ablation evaluation

The quantities on which the deployment claim rests are the accuracy against a human labelling of the central row and the comparisons that interpret it. A subset of 150 frames was hand-labelled by one annotator as a representative sample, stratified across the two runs and the vegetation tertiles with even temporal spacing. Because the raw front end produces large heading excursions on a sizeable fraction of frames, this representative subset already contains 62 such frames, enough to test the temporal ablation without enriching the sample for difficulty. For each labelled frame the annotator marks the central navigation row, and the ground-truth (GT) line and its look-ahead tangent are derived in the same $x = P(y)$ parameterisation as the estimator. Because the labelling was performed by a single annotator, an inter-annotator agreement floor on the smallest meaningful error is not available, and establishing one is left to future work; the errors below are therefore read against a single human reference.

Accuracy against ground truth. Table 5 reports the heading and cross-track error of the estimator against the labelled subset, for the raw per-frame output and for the fused output after the temporal stage. The fused heading error has a median of 5.0° (mean 6.3° , 90th percentile 13.2°), and three

quarters of the labelled frames fall within 10° . The fused cross-track error has a median of 10 px, which is about 1 cm through the calibrated homography. The error distribution has a heavy tail: a minority of frames in which the Excess-Green front end locks onto a neighbouring row rather than the central one raise the cross-track mean to 18 px and the 90th percentile to 43 px (≈ 4.6 cm). These are the cases in which a vegetation index, which has no notion of which plants belong to the crop, is least reliable, and the trained-detector baseline below confirms they are where a learned detector helps most.

Table 5: Accuracy against the 150-frame human-labelled subset (one annotator), for the raw per-frame estimate and the fused estimate after the temporal stage. Errors are absolute differences from ground truth, summarised by median, mean, and 90th percentile.

Error vs. GT	Median	Mean	p90
Heading, raw ($^\circ$)	6.3	9.0	19.4
Heading, fused ($^\circ$)	5.0	6.3	13.2
Cross-track, raw (px)	11	34	74
Cross-track, fused (px)	10	18	43
Cross-track, fused (cm)	1.1	2.0	4.6

Temporal-stage ablation. The temporal stage improves accuracy, and not only smoothness. On the 62 labelled frames on which the max-jump gate fired, the raw heading error has a median of 11.8° against ground truth, which the temporal stage reduces to 8.1° ; the effect replicates across both field runs taken separately (gated heading error $12.7^\circ \rightarrow 8.0^\circ$ and $11.8^\circ \rightarrow 8.3^\circ$). The gate is also selective, in that the mean raw heading error is 14.4° on the gated frames against 5.1° on the rest, so the gate fires on genuinely poor frames rather than rejecting good ones. This is the non-circular form of the temporal result: unlike the jump counts of Section 4.3, the comparison is against an external reference on exactly the frames the stage acts upon. The two front-end and temporal additions that target the wrong-row tail are consistent with this picture. The seed-corridor width is

the one perception hyperparameter selected with reference to this labelled subset: sweeping it, the large cross-track errors (> 100 px) fall from 23 (no corridor) to 18 (± 220 mm) to 13 (± 180 mm), while ± 150 mm over-tightens and worsens the cross-track tail, so ± 180 mm is adopted; because this touches the evaluation subset, choosing it on a disjoint set is left to future work. Stabilising the cross-track signal through the temporal stage, rather than reporting it per frame, then reduced its 90th-percentile error from 74 px (raw) to 43 px (fused).

Fit-arm comparison. Replacing the pipeline’s Huber IRLS fit with the two standard alternatives on the same anchors, each passed through the same fit wrapper so that only the loss differs, changes the result little (Table 6). The Huber and equal-weight least-squares fits are statistically indistinguishable: the paired per-frame difference has a median of 0.0° in heading and 0.0 px in cross-track, each fit is the lower of the two on about a third of frames, and the small gaps in the aggregate mean and tail are not a systematic advantage of either. RANSAC is the weakest of the three (7.2° median heading against 6.3°), because its hard inlier threshold discards anchors the other fits use. On these corridor-restricted anchors the fit choice is therefore immaterial within the least-squares family, which is consistent with the temporal stage rather than the fit being the robustness lever; the Huber fit matches equal-weight least squares here while retaining robustness against outlier anchors when they occur.

Table 6: Fit-arm comparison on the labelled subset: per-frame error of the pipeline’s Huber IRLS fit and the two standard alternatives, on identical anchors through the same fit wrapper, so that only the loss differs.

Fit	Heading err ($^\circ$)			Cross-track err (px)		
	med	mean	p90	med	mean	p90
Huber IRLS (proposed)	6.3	9.0	19.4	11	34	74
Equal-weight LS	6.4	9.2	20.8	10	34	77
RANSAC	7.2	10.5	25.3	12	34	90

Trained-detector baseline. A trained YOLOv5 cabbage detector provides the comparison baseline: its detections are filtered by a ± 220 mm safety corridor and the same curve is fit to them (Table 7), so the two methods share the downstream fit and differ in their perception front end. The headline comparison is between the two methods as each is constituted: the complete proposed pipeline, with its temporal stage, against the trained detector. The trained detector has the lower point estimate on every aggregate—heading median 4.2° against 5.0° , mean 6.1 against 6.3 , 90th percentile 11.9 against 13.2 , and a uniformly lower cross-track—and it is decisively better on the wrong-row tail. The per-frame paired difference is, however, small relative to the sample’s resolution: a median of 0.2° with a run-clustered 95% bootstrap interval of $[-0.5, 1.2]^\circ$ (cross-track 1.1 px, $[-0.9, 3.1]$). An advantage of up to about 1.2° therefore cannot be excluded at $n = 150$; the gap is not resolvable here rather than demonstrably zero. The detector’s decisive advantage is on the 13 frames where the Excess-Green tracker locks onto a neighbouring row: there it localises the central row to a median cross-track error of 13 px against the raw tracker’s 201 px, better on every one of those frames. Two checks bound the comparison. Giving the detector the identical temporal stage, reported as an ablation, barely moves it ($4.2^\circ \rightarrow 4.1^\circ$, cross-track 90th percentile unchanged at 32 px) and the matched paired difference is still unresolved (0.5° , $[-0.4, 1.2]$), so the small gap does not rest on smoothing one side and not the other. And stratifying by how far the ground-truth row sits off-centre shows the gap does not widen into the most off-centre tercile ($|e^{\text{GT}}| > 39$ px: 6.9° proposed against 5.6° detector). The detector-free pipeline thus comes within the sample’s resolution of the trained detector in the typical case, at no labelling or GPU cost, while remaining behind it in point estimate and on the worst-case wrong-row frames.

Table 7: Trained-detector baseline. The headline comparison is method-versus-method: the complete proposed pipeline (ExG + temporal) against a trained YOLOv5 detector, whose detections pass through a safety corridor and the same curve fit, so the two methods share the downstream fit and differ in their perception front end. The YOLOv5 + temporal row is an ablation, included only to show the temporal stage barely moves the strong detector.

Method	Heading err (°)			Cross-track err (px)		
	med	mean	p90	med	mean	p90
ExG, raw	6.3	9.0	19.4	11	34	74
ExG + temporal (proposed)	5.0	6.3	13.2	10	18	43
YOLOv5 (baseline)	4.2	6.1	11.9	9	15	32
+ temporal (ablation)	4.1	5.4	11.7	9	14	32

4.5. Front-end, index, and temporal baselines

The fit-arm comparison varied the curve fit; we further varied the perception front end, the vegetation index, and the temporal filter, each through the same downstream and scored on the labelled subset (Table 8). Two patterns emerge. First, the temporal stage lowers the fused heading error below the raw per-frame error for every usable front end and index: the ExG tracker improves from 6.3° to 5.0°, ExGR from 6.0° to 4.0°, CIVE from 5.7° to 5.0°, and the centroid-per-band tracker from 7.8° to 5.2°, so the improvement is a property of the temporal stage rather than of any one front end. Second, the column-peak trackers cluster at a fused median of 4–5° whichever standard index feeds them, so the index choice is immaterial: the sub-degree differences among the standard indices (ExGR nominally lowest at 4.0°, ExG and CIVE at 5.0°) sit within the paired bootstrap interval and are not ranked. The two genuinely different front ends fare worse: the periodic-row tracker reaches 8.4°, and the Hough-line detector is the clear outlier, emitting a guidance line on only 59% of frames and reaching 23.9°, because the discrete, gapped cabbage plants do not form the continuous edges a line detector relies on. The temporal stage does not rescue Hough – its fused error is no lower than its raw error – because its failures are systematic rather than the transient spikes the median and gate

are designed to remove. Across the usable front ends and indices the temporal stage is therefore what lowers the error, and the perception choice is secondary.

Table 8: Front-end and vegetation-index baselines, each run through the same fit and temporal stage and scored on the labelled subset: emission rate and fused heading error. The first block swaps the vegetation index in the proposed column-peak tracker; the second swaps the front end itself.

Method	Emit	Fused heading err (°)		
		med	mean	p90
ExG column-peak (proposed)	100%	5.0	6.3	13.2
index ExGR	100%	4.0	5.7	13.6
index CIVE	100%	5.0	6.6	14.7
index NDI	100%	7.1	7.5	12.7
Centroid-per-band	100%	5.2	6.4	13.9
Periodic-row	100%	8.4	8.1	13.8
Hough lines	59%	23.9	21.9	39.9

The temporal stage is also preferable to a more elaborate filter. Replacing the proposed median, exponential moving average, and jump gate with a constant-velocity Kalman filter on the same raw ExG heading stream gives a fused median heading error of 5.8° , between the raw 6.3° and the proposed 5.0° ; the simple stage, whose complexity is matched to the 1 fps regime, is therefore sufficient and marginally better here.

The two principal limitations of these accuracy figures are the single annotator, for which an inter-annotator agreement floor is left to future work, and the residual wrong-row tail of the vegetation-index front end that the detector comparison identifies as the main accuracy gap.

5. Discussion

This is an applied contribution: an in-field evaluation of a detector-free, training-free guidance pipeline of stock parts on a cabbage robot, not a new method. The discussion takes up why the

temporal stage rather than the perception front end carries the robustness, the failure modes of the vegetation-index front end and the cost of avoiding a trained detector, and the limits of the evidence.

5.1. Why the temporal stage is the robustness lever

A steering controller integrates whatever heading it is handed, so what matters is not the mean heading error but the rate of *large* errors: one frame in which the Excess-Green front end latches onto a weed, a gap, or the wrong row can swing the look-ahead tangent by tens of degrees, and such a spike, if acted on, could drive the robot into the crop. Suppressing such spikes is the temporal stage’s job; the raw-versus-fused jump counts of Table 4 confirm that the gate does so by construction, and the test that matters is error against ground truth on the frames it acts upon.

The Excess-Green tracker is a single-frame estimator that commits, per frame, to the strongest column of vegetation contrast with no memory of the last few seconds; any stock single-frame front end shares this weakness, because the errors come from scene content (a weed band, a canopy gap, an adjacent row) rather than from the estimator. The temporal stage – a median that rejects isolated contaminated frames, an EMA that limits how fast the target moves, and a jump gate at the vehicle’s kinematic limit – is what converts an occasionally-wrong per-frame estimate into a heading a controller can integrate. The labelled ablation bears this out: on the gate-suppressed frames it lowers the median heading error against GT from 11.8° to 8.1° , and the gate fires selectively, with a mean raw error of 14.4° on the frames it holds against 5.1° on the rest (Section 4.4). The fit-arm and front-end/index baselines point the same way: the fit choice is secondary and the temporal stage improves every front end and index tried, so within the set we tested the temporal stage, not the perception choice, is the lever, although generality beyond that set is unproven. One caveat is unmeasured: the gate’s selectivity shows the frames it holds carry larger errors on average, but whether it ever vetoes a *correct* sharp correction – at a headland, a bend, or a genuine drift – is the

false-reject rate that a dedicated trial with annotated sharp-correction events would need to establish before the gate can be called a safety mechanism.

5.2. Failure modes of the Excess-Green front end

Because the temporal stage can only stabilise what the front end gives it, the failure modes of the Excess-Green tracker bound the system. They are the familiar limits of vegetation-index row sensing [6, 9, 10], and they bear on cabbage forward-view imagery in specific ways. *Low vegetation contrast*: the Excess-Green index separates green canopy from soil by colour, so at early growth stages, over wet or dark soil that is itself greenish, or under flat overcast light, the column-peak signal is weak and the tracked centre wanders. *Weeds*: inter-row weeds are also green and also produce Excess-Green columns, so a weed band between rows can out-peak the crop row and pull the tracker off-centre, a content failure that no amount of robust fitting on a single frame can detect. *Canopy closure*: as cabbage matures and neighbouring rows touch, the inter-row soil gap that defines the column structure closes, the peaks merge, and the central-row column becomes ambiguous; the tracker may then follow the seam between two rows rather than a row centre. *Gaps and missing plants*: a stretch of missing plants removes vegetation contrast locally, so the look-ahead portion of the curve is fit on little or no support and the tangent becomes unstable exactly where heading is most sensitive. In all four cases the front end produces a confident-looking but wrong per-frame line, which is why the temporal stage matters. It is also why the temporal stage's protection is bounded by its false-reject rate: a sustained content failure such as a long weed band or a closed-canopy stretch is not a transient spike, and a median-plus-gate designed for spikes will eventually track it. The ground safety corridor and the temporal stabilisation of the cross-track reduce this wrong-row tail but do not remove it; the residual is the content failure a vegetation index cannot resolve, and it is exactly where the trained detector is decisively more accurate (Section 4.4).

5.3. Limitations

Several boundaries limit how far these conclusions generalise. This study measures guidance accuracy against ground truth; it does not close the steering loop. The smoothness of the fused heading trace is an accuracy property and not a measured closed-loop tracking result, and the temporal parameters (window length, EMA factor, jump threshold) interact with controller and vehicle dynamics that an accuracy evaluation does not exercise. Integrating the guidance into the steering loop and measuring the realised intervention rate and tracking error with wheel odometry is the clear next step. The accuracy itself is, however, robust to how far the row sits off-centre: stratifying every trained-detector comparison by $|e^{\text{GT}}|$ shows the small pipeline-versus-detector gap does not widen into the most off-centre tercile (Section 4.4).

The metric figures rest on a measured camera pose and a flat-ground assumption. The heading θ is reported in degrees, which is controller-relevant and scale-free, and the cross-track offset e is reported in pixels, as a fraction of half the image width, and in centimetres. The centimetre conversion uses an image-to-ground homography built from the robot’s own measured camera height (65 cm) and tilt (55° from the vertical), with the camera intrinsics reused from a checkerboard calibration of the same camera. A flat-ground homography still inflates metric error most at the far look-ahead distance, precisely where guidance is most sensitive, so the centimetre figures away from the near row carry that planar-ground caveat. A full on-robot checkerboard calibration, including the intrinsics, would further tighten the metric mapping.

The scope is narrow: two runs, one crop, one camera pose. The evidence is two real cabbage runs (~ 20 and ~ 14 minutes, 1482 curated cabbage frames at 1 fps, 640×480), from one robot, one crop, and one forward-view camera pose. This is a pure test set, since there is no training stage anywhere in the pipeline and no learned weights can leak from these frames into the method, but

it is also a small, single-condition sample. With $n = 2$ runs the across-field, across-season, and across-platform variability cannot be characterised, and the descriptive front end statistics already differ markedly between the two runs (raw $> 10^\circ$ jumps in 14.1% versus 28.3% of transitions, that is 114/810 and 189/669), which is itself a caution against generalising from two clips.

Finally, accuracy is measured against a single annotator, so an inter-annotator agreement floor on the smallest meaningful error is not established; a second annotator is left to future work. To avoid over-reading differences that may sit within label noise, comparisons whose paired bootstrap interval spans zero – the sub-degree detector gap (reported as not resolvable at this sample size, with the detector still ahead in point estimate) and the Huber-versus-least-squares fit difference (median zero) – are reported as unresolved rather than ranked.

The trained-detector baseline is also a single reference. Its detections are filtered by a ± 220 mm corridor while the pipeline seeds inside a ± 180 mm band, so the wrong-row comparison is not corridor-matched, and the detector’s training set, growth-stage coverage, and held-out accuracy are not characterised here. The comparison should therefore be read as against one trained detector at one operating point, not against trained detectors in general; a corridor-matched evaluation and a documented detector are left to future work.

6. Conclusions

On a low-cost field robot without RTK-GPS, the quantities a steering controller actually consumes are a heading and a cross-track offset, not a detection score [4, 5]. Adopting that interface, this work examines what crop-row guidance accuracy is achievable with a pipeline that has no trained detector and no learned weights of any kind. The guidance estimator is assembled entirely from well-understood stock parts. An Excess-Green index [9, 10] feeds a column-peak central-row

tracker; a robust quadratic fit $x = P(y)$ by Huber IRLS ($\delta = 1.345$, scale $1.4826 \cdot \text{MAD}$) models the guidance line [6, 11]; the heading θ is taken as the look-ahead tangent to P and the cross-track offset e at the vehicle reference row; and a temporal stage of a sliding-window median, an exponential moving average, and a physical max-jump gate smooths both signals [12]. All components are established. The study offers an in-field validation of what this stock composition does on a real cabbage robot, rather than a new method.

Against a 150-frame human-labelled subset the detector-free pipeline reaches a median heading error of 5.0° and a cross-track of about 1 cm, with a guidance line emitted on 99.9% of frames. Two findings stand out. The temporal stage, not the perception front end, carries the robustness: it lowers heading error on the gate-suppressed frames in both runs, fires selectively on genuinely poor frames, and improves every front end and index tried, while the fit choice is secondary (Section 4). And against a trained YOLOv5 detector baseline the detector-free pipeline trails by a median heading gap of only 0.2° that is not resolvable at $n = 150$ (95% interval $[-0.5, 1.2]$, and still unresolved when the detector is given the same temporal stage) at no labelling or GPU cost; the detector stays uniformly ahead in point estimate and decisively better on the wrong-row frames where a vegetation index cannot separate crop from weed – the measured cost of going detector-free.

The scope is one robot, one crop, and one camera pose, at 1 fps; the claims extend no further. The dataset and code will be made publicly available upon publication.

6.1. Future work

Four steps follow from the boundaries above. An on-robot checkerboard calibration would replace the reused intrinsics and flat-ground homography with one measured on the cabbage robot. Closing the steering loop from the pipeline’s output, with wheel odometry, would turn the measured guidance accuracy into a realised intervention rate and expose how controller and terrain dynamics interact

with the gate [25]. A second annotator would establish the inter-annotator agreement floor this single-annotator study lacks. And extending the dataset to further row crops would test whether the temporal stage remains the robustness lever under genuine distribution shift.

Data and Code Availability

The forward-view cabbage-robot guidance dataset (two field runs, 1482 forward-view cabbage frames at 640×480 screened from 1 fps sampling, with the per-frame guidance logs), the 150-frame hand-labelled ground-truth subset, and the detector-free guidance code will be made publicly available upon publication. A repository link and archival DOI will be provided in the camera-ready version.

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Contributions

Conceptualization, investigation, and formal analysis: Manh V. Hoang, Nam Do and Thang M. Pham; methodology, validation: Manh V. Hoang and Thang M. Pham; software and writing: Manh V. Hoang. All authors were involved in its revision. All authors read and approved the final manuscript.

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